Krishiv Patel

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Education

Rutgers University, School of Engineering — Piscataway, NJ
B.S. Aerospace Engineering, Minor in Mathematics, Energy Systems Certificate — Expected May 2027

- GPA 3.44/4.00 | Dean's List: 4 semesters
- James J. Slade Scholar: graduate research program, 6 credits of graduate independent research
- Honors: Francisco Diaz-Masso Engineering Scholarship, NASA Space Grant Scholarship, Rutgers School of Engineering Scholarship

Experience

Rutgers Department of Mechanical & Aerospace Engineering — Piscataway, NJ
NASA Space Grant Fellow (2026–27) | Research Lead, Design & CFD (from Undergraduate Researcher) — Mar 2025 – Present

- Lead design on a $42K pulsed-air actuator replacing fixed nacelle chines with deployable active flow control, 150–200 m/s jets at 10–1000 Hz
- Re-architected after a manufacturability review with Calspan and GE engineers: a localized plenum replaced the external compressor feed, decoupling pressure from modulation; aerodynamics reviewed by two former Boeing engineers including a Technical Fellow
- Own CFD (ANSYS Fluent, CFX) and parametric CAD (SolidWorks); tolerance to ASME Y14.5-2018, informed by hands-on fabrication certification

Rutgers Rocket Propulsion Lab — Piscataway, NJ
Design & Aerodynamics, 750 lbf Lab Engine Program — Mar 2025 – Present

- Size coolant channels and run flow analysis for regenerative cooling on the lab's 750 lbf pressure-fed N2O/IPA engine, which hot fired in May 2026; resulting design changes cut chamber wall temperature 3%
- Size injector plate orifices to maintain thermal separation between regenerative coolant passages and the solid rocket motor igniter, translating hand calculations into geometry revisions

NASA New Jersey Space Grant Consortium – Rutgers University — Piscataway, NJ
Experimental Aerodynamics Research Intern — May 2026 – Aug 2026

- Directed five Governor's School of Engineering & Technology scholars, from the top 64 high school engineers in New Jersey, on a NASA-funded study open only to James J. Slade Scholars
- Built the test setup, calibration procedure, and data reduction workflow; ran CAD, CFD, 3D printing, and wind tunnel testing across 51 biplane configurations at Re = 20,100
- Peak L/D 9.12 across four trials vs 4.21 monoplane; corresponding author on both resulting papers

Rutgers Department of Mechanical & Aerospace Engineering — Piscataway, NJ
Wind Tunnel Lab Developer — Aug 2026 – Present

- Co-developing the wind tunnel module for the Rutgers MAE Senior Lab, a required course for all Aerospace and Mechanical seniors
- Redesigned and reprinted the airfoil model set for repeated student use, and documented the test's limitations so students can run it without supervision

Publications

Experimental Investigation of Airfoil Aerodynamics and Flow Control Concepts: Mono and Biplane Configurations

- IEEE MIT Undergraduate Research Technology Conference, presented Oct 2026 | first author; co-corresponding with Dr. DeMauro
- GSET Research Journal, published July 2026 | sole corresponding author

Projects

750 lbf IPA/N2O Bipropellant Rocket Engine *(Head of Design, 8-person team, Rutgers Senior Design)* — May 2026 – Present

- Own injector plate design: converted element geometry from coaxial swirl to shear coaxial for machinability and uniform propellant distribution, and redesigned the igniter interface as a replaceable canister so the team can iterate between hot fires
- Diagnosed prior-generation engine: 50 lbf measured against a 750 lbf target, traced to unsustained chamber ignition driving combustion downstream of the throat with no choked flow; redesigning igniter integration to establish sustained in-chamber combustion
- Own vehicle integration and the static test stand for cold-flow characterization and hot fire, including feed system design and high-pressure procedures

Spacecraft Thermal Subsystem *(NASA L'SPACE Mission Concept Academy)* — Aug 2026 – Present

- Own the thermal subsystem for a robotic lunar south pole payload exploring permanently shadowed regions near 40 K: thermal control architecture, environment definition, and component temperature margins
- Designing to a 250 kg, $250M envelope with no RTG power, through NASA life-cycle gate reviews to a Preliminary Design Review assessed by NASA subject matter experts

Hypersonic Nose Cone Aerothermal Analysis *(University Consortium for Applied Hypersonics)* — Jul 2025 – Dec 2025

- Analyzed multiple nose cone geometries at hypersonic conditions to characterize shockwave structure and aerodynamic heating, splitting flow solution in ANSYS Fluent with conduction analysis in ANSYS Mechanical (Transient Thermal)
- Conducted blunt vs sharp trade study balancing drag against stagnation heating; integrated an internal honeycomb pattern for conductive heat dissipation

Technical Skills

Analysis: ANSYS Fluent, ANSYS CFX, ANSYS Mechanical (Transient Thermal), XFLR5, MATLAB, Python, LabVIEW; classical lamination theory, composite failure criteria (Hashin, max stress)
Design: SolidWorks, OnShape, ANSYS SpaceClaim, GD&T (ASME Y14.5-2018), Design for Manufacturability
Experimental & Fabrication: Wind tunnel testing and calibration, hot-wire anemometry, smoke flow visualization, Shopbot CNC, laser cutting, FDM 3D printing